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Power amplifier circuit

Abstract

A power amplifier circuit includes a first transistor having the base coupled to an input terminal, the collector coupled to an output terminal, and the emitter coupled to ground, a first bias circuit coupled to the base of the first transistor via the first resistance element, a second transistor having the base coupled to the input terminal, the collector coupled to the output terminal, and the emitter coupled to ground, a second bias circuit coupled to the base of the second transistor via the second resistance element, and a first impedance circuit having a first end coupled between the base of the first transistor and the input terminal and a second end coupled between the first bias circuit and the first resistance element and being configured to be opened for a direct-current component and to be closed for an alternating-current component.

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Patent No.	Application Date	Country	CPC
4027349	12/2006	JP	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION

(1) This application claims priority from Japanese Patent Application No. 2021-021652 filed on Feb. 15, 2021. The content of this application is incorporated herein by reference in its entirety.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

(2) The present disclosure relates to a power amplifier circuit.

2. Description of the Related Art

(3) In some types of power amplifiers, base bias voltage is supplied through a resistance element to a common-emitter bipolar transistor (refer to, for example, Japanese Patent No. 4027349 (Patent Document 1)).

BRIEF SUMMARY OF THE DISCLOSURE

(4) The power amplifier described in Patent Document 1 includes a first impedance circuit and a second impedance circuit. The first impedance circuit has one terminal coupled to a base terminal of a common-emitter bipolar transistor and the other terminal coupled to a signal input terminal.

The second impedance circuit has one terminal coupled to the signal input terminal and the other terminal coupled to a base-bias-voltage feeding terminal. At least either the first impedance circuit or the second impedance circuit is opened for direct-current components; both the first impedance circuit and the second impedance circuit are closed for alternating-current components.

(5) Because the first impedance circuit and the second impedance circuit are provided in the transfer path for alternating current signals, increases in voltage drop at the resistance element are suppressed. This reduces gain compression, and as a result, the linearity of the amplification characteristic can be maintained. This means that it is possible to improve the distortion (harmonic wave) characteristic.

(6) Of power amplifiers used in present mobile communication devices such as mobile phones, some types of power amplifiers increase or decrease their output level to different levels in accordance with the distance between the mobile communication device and the base stations. For such power amplifiers, a demand exists for a technology for suppressing the degradation of the distortion characteristic when the output level is increased or decreased to different levels (changed between different levels).

(7) The present disclosure has been made in consideration of the above circumstances, and an object thereof is to provide a power amplifier circuit capable of suppressing the degradation of the distortion characteristic when its output level is changed between different levels.

(8) A power amplifier circuit according to an aspect of the present disclosure includes a first transistor having the base coupled to an input terminal, the collector coupled to an output terminal, and the emitter coupled to ground, a first resistance element, a first bias circuit coupled to the base of the first transistor via the first resistance element and configured to supply a bias to the first transistor, a second transistor having the base coupled to the input terminal, the collector coupled to the output terminal, and the emitter coupled to ground, a second resistance element, a second bias circuit coupled to the base of the second transistor via the second resistance element and configured to supply a bias to the second transistor, and a first impedance circuit having a first end coupled between the base of the first transistor and the input terminal and a second end coupled between the first bias circuit and the first resistance element and being configured to be opened for a direct-current component and to be closed for an alternating-current component.

(9) The present disclosure can provide the power amplifier circuit capable of suppressing the degradation of the distortion characteristic when its output level is changed between different levels.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

- (1) FIG. 1 is a circuit diagram of a power amplifier circuit;
- (2) FIG. 2 is a circuit diagram of a bias circuit;
- (3) FIG. 3 is a circuit diagram of a high-frequency amplifier circuit of a reference example;
- (4) FIG. 4 depicts input-output characteristics of the high-frequency amplifier circuit illustrated in FIG. 3;
- (5) FIG. 5 is a circuit diagram of a power amplifier circuit;
- (6) FIG. 6 is a schematic diagram in which a wire of the power amplifier circuit is drawn as a straight line;
- (7) FIG. 7 is a schematic diagram in which a wire of the power amplifier circuit is drawn as a straight line;
- (8) FIG. 8 is a circuit diagram of a power amplifier circuit;
- (9) FIG. 9 is a circuit diagram of a power amplifier circuit;
- (10) FIG. 10 is a circuit diagram of a power amplifier circuit;

- (11) FIG. **11** is a circuit diagram of a power amplifier circuit;
- (12) FIG. **12** is a circuit diagram of a power amplifier circuit;
- (13) FIG. **13** is a circuit diagram of a power amplifier circuit;
- (14) FIG. **14** is a circuit diagram of a power amplifier circuit; and
- (15) FIG. **15** is a circuit diagram of a power amplifier circuit.

DETAILED DESCRIPTION OF THE DISCLOSURE

(16) Hereinafter, preferred embodiments of the present disclosure will be described in detail with reference to the drawings. It should be noted that the same elements are assigned with the same reference characters, and redundant descriptions are omitted as much as possible.

First Embodiment

(17) The following describes a power amplifier circuit according to a first embodiment. FIG. **1** is a circuit diagram of a power amplifier circuit **11**. As illustrated in FIG. **1**, the power amplifier circuit **11** according to the first embodiment is a circuit configured to amplify an input signal (radio-frequency signal) RFin and outputs an output signal (amplified signal) RFout.

(18) The power amplifier circuit **11** includes a bias circuit **301** (first bias circuit), a bias circuit **302** (second bias circuit), amplifiers **401** and **402**, and an impedance circuit **501** (first impedance circuit). The amplifier **401** includes a unit transistor **101** (first unit). The unit transistor **101** includes a transistor element **51** (first transistor), a capacitor **52** (impedance circuit), and a resistance element **53** (first resistance element). The amplifier **402** includes the unit transistor **201** (second unit). The unit transistor **201** includes a transistor element **61** (second transistor), a capacitor **62** (impedance circuit), and a resistance element **63** (second resistance element). The elements constituting the power amplifier circuit **11** are formed on the same semiconductor substrate. The elements constituting the power amplifier circuit **11** may be formed on a plurality of semiconductor substrates. The unit transistor denotes a configuration formed by combining together a transistor element, a capacitor, and a resistance element.

(19) In the description of the present embodiment, the elements including the transistor elements **51** and **61** are configured by bipolar transistors such as heterojunction bipolar transistors (HBTs). The transistor elements **51** and **61** may be configured by other kinds of transistors such as metal-oxide-semiconductor field-effect Transistors (MOSFETs). In this case, base, collector, and emitter are respectively regarded as gate, drain, and source.

(20) The transistor element **51** of the amplifier **401** has the base coupled to an input terminal **31** via the capacitor **52**, the collector coupled to a supply-voltage feeding node N1 for feeding a supply voltage VCC1 and an output terminal **32**, and the emitter coupled to the ground.

(21) The resistance element **53** has a first end coupled to the bias circuit **301** and a second end coupled to the base of the transistor element **51**. The capacitor **52** has a first end coupled to the input terminal **31** and a second end coupled to the base of the transistor element **51**.

(22) The bias circuit **301** is coupled to the base of the transistor element **51** via the resistance element **53**. The bias circuit **301** supplies a bias to the transistor element **51**. Details of the bias circuit **301** will be described later.

(23) The transistor element **61** of the amplifier **402** has the base coupled to the input terminal **31** via the capacitor **62**, the collector coupled to the supply-voltage feeding node N1 and the output terminal **32**, and the emitter coupled to the ground.

(24) The resistance element **63** has a first end coupled to the bias circuit **302** and a second end coupled to the base of the transistor element **61**. The capacitor **62** has a first end coupled to a node provided in a wire W1 (first wire) connecting the input terminal **31** and the first end of the capacitor **52**, and a second end coupled to the base of the transistor element **61**.

(25) The bias circuit **302** is coupled to the base of the transistor element **61** via the resistance element **63**. The bias circuit **302** supplies a bias to the transistor element **61**. Details of the bias circuit **302** will be described later.

(26) The impedance circuit **501** has a first end coupled between the base of the transistor element

51 and the input terminal **31** and a second end coupled between the bias circuit **301** and the resistance element **53**. The impedance circuit **501** is opened for direct-current components and closed for alternating-current components.

(27) In the present embodiment, the first end of the impedance circuit **501** is coupled to a node **N11** (first node) provided in the wire **W1**. The second end of the impedance circuit **501** is coupled to a node **N12** provided in a wire **W4** connecting the bias circuit **301** and the first end of the resistance element **53**.

(28) FIG. 2 is a circuit diagram of the bias circuit **301**. As illustrated in FIG. 2, the bias circuit **301** includes transistors **301a**, **301b**, and **301c**, a capacitor **301d**, and a resistance element **301e**. The bias circuit **302** has the same configuration as the bias circuit **301**; the bias circuit **301** is described here as the representative, and the description of the bias circuit **302** will be omitted as much as possible. The bias circuit may be configured in any manner other than the manner of the circuit described above.

(29) The bias circuit **301** supplies a bias to the base of the transistor element **51** by using the transistor **301a** emitter-follower coupled to the base of the transistor element **51**.

(30) Specifically, the transistor **301a** has the collector coupled to a battery-voltage feeding node **N2** for feeding a battery voltage V_{bat} , the base, and the emitter coupled to the first end of the resistance element **53** and the second end of the impedance circuit **501**.

(31) The transistors **301b** and **301c**, the capacitor **301d**, and the resistance element **301e** supply a bias of a predetermined level of voltage to the base of the transistor **301a**.

(32) Specifically, the resistance element **301e** has a first end coupled to a current feeding node **N3** and a second end. Under the control of a bias control circuit (not illustrated in the drawing) in a preceding stage, a control bias current I_{cb} for controlling the amplification of the transistor element **51** or **61** is fed from the current feeding node **N3**. Details of the control of the bias control circuit will be described later. It should be noted that voltage may be supplied to the current feeding node **N3**.

(33) The transistor **301b** has the collector coupled to the second end of the resistance element **301e** and the base of the transistor **301a**, the base coupled to the collector, and the emitter. In the following description, the coupling of the collector and base of a transistor may be referred to as diode coupling. The diode-coupled transistor may be replaced with a diode element.

(34) The transistor **301c** is formed by diode coupling. The transistor **301c** has the collector coupled to the emitter of the transistor **301b** and the emitter coupled to the ground. Because both the transistors **301b** and **301c** function as diodes, a magnitude of voltage drop corresponding to two diodes occurs in a path between the collector and emitter of the transistor **301b** and a path between the collector and emitter of the transistor **301c**. This means that the voltage at the collector of the transistor **301b** is higher by the magnitude of voltage drop corresponding to two diodes than the reference voltage at the ground. This voltage is supplied to the base of the transistor **301a**.

(35) The capacitor **301d** has a first end coupled to the second end of the resistance element **301e** and a second end coupled to the ground. The capacitor **301d** stabilizes the voltage at the collector of the transistor **301b**.

(36) Power Mode

(37) As illustrated in FIGS. 1 and 2, the power amplifier circuit **11** operates in, for example, two modes of a high-power mode and a low-power mode.

(38) For example, when the power amplifier circuit **11** operates in the high-power mode, the control bias current I_{cb} is supplied from the current feeding node **N3** in the bias circuits **301** and **302** under the control of the bias control circuit (not illustrated in the drawing).

(39) At this time, both the transistor elements **51** and **61** provide amplification, and thus, the output signal RF_{out} of relatively high output power is outputted from the output terminal **32**.

(40) By contrast, when the power amplifier circuit **11** operates in the low-power mode, under the control of the bias control circuit (not illustrated in the drawing), the control bias current I_{cb} is

supplied from the current feeding node N3 in the bias circuit 302, while the control bias current I_{cb} is not supplied from the current feeding node N3 in the bias circuit 301.

(41) At this time, the transistor element 61 provides amplification, but the transistor element 51 does not provide amplification; accordingly, the output signal RF_{out} of relatively low output power is outputted from the output terminal 32.

(42) Effects on Thermal Runaway

(43) Between the bias circuit 301 and the base of the transistor element 51, the impedance circuit 501 is opened for direct-current components. As a result, the direct-current component of a bias current I_{b1} flowing from the bias circuit 301 to the base of the transistor element 51 passes through only the resistance element 53.

(44) For example, when the temperature of the transistor element 51 increases under the influence of, for example, amplification operation, the on-voltage of the transistor element 51 lowers, the direct-current component of the bias current I_{b1} increases, and a current I_{c1} flowing from the collector of the transistor element 51 to the emitter of the transistor element 51 increases. At this time, in response to the increase of the direct-current component of the bias current I_{b1} , voltage drop across the resistance element 53 also increases. This can consequently lower the electric potential at the base of the transistor element 51 relative to the ground. As a result, it is possible to suppress increases of the direct-current component of the bias current I_{b1} and also suppress increases of the current I_{c1} . Accordingly, it is possible to reduce the likelihood of thermal runaway in the transistor element 51.

(45) Also, in the amplifier 402, voltage drop across the resistance element 63 increases in response to increases in the temperature of the transistor element 61. As a result, it is possible to suppress increases in the direct-current component of a bias current I_{b2} flowing from the bias circuit 302 to the base of the transistor element 61 and also suppress increases in a current I_{c2} flowing from the collector of the transistor element 61 to the emitter of the transistor element 61. Accordingly, it is possible to reduce the likelihood of thermal runaway in the transistor element 61.

(46) Effects on Gain Compression

(47) FIG. 3 is a circuit diagram of a high-frequency amplifier circuit of a reference example. FIG. 4 depicts input-output characteristics of the high-frequency amplifier circuit illustrated in FIG. 3. The circuit diagram illustrated in FIG. 3 and the input-output characteristics depicted in FIG. 4 are described in, for example, Japanese Unexamined Patent Application Publication No. 9-260964.

(48) As illustrated in FIGS. 3 and 4, a base-emitter direct-current voltage V_{BE} lowers as an input-signal power P_{in} increases. The base-emitter direct-current voltage V_{BE} corresponds to the electric potential at the base of a bipolar transistor $Tr1$.

(49) The direct current flowing from the base of the bipolar transistor $Tr1$ to the emitter of the bipolar transistor $Tr1$ flows from the base-bias-voltage feeding terminal VB for the bipolar transistor $Tr1$ via a resistance element R1, the base of the bipolar transistor $Tr1$, and the emitter of the bipolar transistor $Tr1$, to the ground.

(50) As described above, the base-emitter direct-current voltage V_{BE} lowers as the input-signal power P_{in} increases; and accordingly, the electric potential at the base of the bipolar transistor $Tr1$ lowers as the input-signal power P_{in} increases. This means that voltage drop across the resistance element R1 increases. This indicates that the base-bias point of the bipolar transistor $Tr1a$ is effectually shifted from class A to class B.

(51) The electric potential at the base of the bipolar transistor $Tr1$ fluctuates similarly to alternating-current voltage in response to high-frequency input signals, and as a result, alternating current flows between the base-bias-voltage feeding terminal VB and the base of the bipolar transistor $Tr1$.

(52) When the bias point is shifted to class B, no base current flows for a given period. This hinders the flow of the alternating-current component (hereinafter also referred to as alternating current I_{Ab}) of the base current flowing between the base-bias-voltage feeding terminal VB and the base

of the bipolar transistor **Tr1**, and consequently, the alternating current **IAb** stops increasing in proportion to the input-signal power **Pin**.

(53) When the alternating current of a high-frequency input signal is **IAS**, and the alternating current inputted to the base of the bipolar transistor **Tr1** is **IAi**, $IAS + IAb \approx IAi$.

(54) The alternating current **IAS** increases in proportion to increases in the input-signal power **Pin**. By contrast, when the input-signal power **Pin** increases, the alternating current **IAb** does not increase in proportion to the input-signal power **Pin** as described above. This means that, when the input-signal power **Pin** increases, the alternating current **IAi** does not increase in proportion to the input-signal power **Pin**, and this results in gain compression in which the output-signal power does not increase in proportion to the input-signal power **Pin**.

(55) To reduce the gain compression, it is effective to set the resistance of the resistance element **R1** to a relatively small value to reduce voltage drop across the resistance element **R1**. Accordingly, the alternating current **IAi** increases in proportion to the input-signal power **Pin**. However, when the resistance of the resistance element **R1** is relatively low, the bipolar transistor **Tr1** operates in a thermally unstable manner.

(56) By contrast, in the power amplifier circuit **11** illustrated in FIG. 1, the impedance circuit **501** and the capacitor **52** are provided in parallel with the resistance element **53**, and thus, the alternating-current component of the bias current **Ib1** is caused to flow into the base of the transistor element **51** through the impedance circuit **501** and the capacitor **52**. As a result, without lowering the resistance of the resistance element **53**, this can achieve the same effect for alternating current as lowering the resistance of the resistance element **53**, and consequently, it is possible to reduce the gain compression. As such, the linearity of the amplification characteristic of the transistor element **51** can be maintained, and the distortion characteristic of the power amplifier circuit **11** is improved.

(57) In the description of the power amplifier circuit **11**, both the transistor elements **51** and **61** provide amplification in the high-power mode, and only the transistor element **61** provides amplification in the low-power mode. This, however, should not be construed in a limiting sense. For example, when the output power of the transistor element **51** is higher than the output power of the transistor element **61**, only the transistor element **51** may provide amplification in the high-power mode, and only the transistor element **61** may provide amplification in the low-power mode. Also, in this case, it is possible to reduce the likelihood of thermal runaway in the transistor elements **51** and **61** and also reduce the gain compression of the transistor element **51** in the high-power mode.

Second Embodiment

(58) The following describes a power amplifier circuit according to a second embodiment. In the second and subsequent embodiments, descriptions of specifics common to the first embodiment are not repeated, and only different points will be explained. In particular, the same effects and advantages achieved by the same configurations are not mentioned in every embodiment.

(59) FIG. 5 is a circuit diagram of a power amplifier circuit **12**. As illustrated in FIG. 5, the power amplifier circuit **12** according to the second embodiment differs from the power amplifier circuit **11** according to the first embodiment in that the impedance circuit **601** is provided between the bias circuit **302** and the input terminal **31**.

(60) When compared to the power amplifier circuit **11** illustrated in FIG. 1, the power amplifier circuit **12** further includes the impedance circuit **601** (second impedance circuit).

(61) The impedance circuit **601** has a first end coupled between the input terminal **31** and the base of the transistor element **61** and a second end coupled between the bias circuit **302** and the resistance element **63**. The impedance circuit **601** is opened for direct-current components and closed for alternating-current components.

(62) In the present embodiment, the first end of the impedance circuit **601** is coupled to a node **N21** (second node) provided in a wire **W2** (second wire) connecting the input terminal **31** and the first

end of the capacitor **62**. The second end of the impedance circuit **601** is coupled to a node **N22** provided in a wire **W5** connecting the bias circuit **302** and the first end of the resistance element **63**.
(63) The impedance circuit **501** includes a capacitor **501a** (first capacitor). Specifically, the capacitor **501a** has a first end coupled to the node **N11** and a second end coupled to the node **N12**.
(64) With this configuration, the capacitor **501a** can easily implement the impedance circuit **501** configured to be opened for direct-current components and closed for alternating-current components.

(65) The impedance circuit **601** includes a capacitor **601a** (second capacitor). Specifically, the capacitor **601a** has a first end coupled to the node **N21** and a second end coupled to the node **N22**.
(66) With this configuration, the capacitor **601a** can easily implement the impedance circuit **601** configured to be opened for direct-current components and closed for alternating-current components.

(67) FIG. **6** is a schematic diagram in which the wire **W1** of the power amplifier circuit **12** is drawn as a straight line. As illustrated in FIG. **6**, a length **L11** of the wire from the input terminal **31** to the node **N11** is more than 0.5 times longer than a length **L10** of the wire from the input terminal **31** to the base of the transistor element **51**.

(68) With this configuration, the alternating-current component of the bias current **Ib1** flowing from the bias circuit **301** to the base of the transistor element **51** can be caused to efficiently flow through the impedance circuit **501** and the capacitor **52** to the base of the transistor element **51**. As a result, it is possible to effectively reduce the gain compression of the amplifier **401** and also suppress the degradation of the distortion characteristic of the amplifier **401**.

(69) FIG. **7** is a schematic diagram in which the wire **W2** of the power amplifier circuit **12** is drawn as a straight line. As illustrated in FIG. **7**, a length **L21** of the wire from the input terminal **31** to the node **N21** is more than 0.5 times longer than a length **L20** of the wire from the input terminal **31** to the base of the transistor element **61**.

(70) With this configuration, the alternating-current component of the bias current **Ib2** flowing from the bias circuit **302** to the base of the transistor element **61** can be caused to efficiently flow through the impedance circuit **601** and the capacitor **62** to the base of the transistor element **61**. As a result, it is possible to effectively reduce the gain compression of the amplifier **402** and also suppress the degradation of the distortion characteristic of the amplifier **402**.

Third Embodiment

(71) The following describes a power amplifier circuit according to a third embodiment. FIG. **8** is a circuit diagram of the power amplifier circuit **13**. As illustrated in FIG. **8**, the power amplifier circuit **13** according to the third embodiment differs from the power amplifier circuit **12** according to the second embodiment in that the amplifier **401** includes a plurality of the unit transistors **101**, and the amplifier **402** includes a plurality of the unit transistors **201**.

(72) When compared to the power amplifier circuit **12** illustrated in FIG. **5**, the power amplifier circuit **13** includes the amplifiers **1401** and **1402** instead of the amplifiers **401** and **402**. The amplifier **1401** includes a number **m** of the unit transistors **101**, where **m** is an integer equal to or greater than 2. In FIG. **8**, the number **m** of the unit transistors **101** are uniquely indicated by reference characters **101** to **10m**. The unit transistors **101** to **10m** each include the transistor element **51**, the capacitor **52**, and the resistance element **53**.

(73) The amplifier **1402** includes a number **n** of the unit transistors **201**, where **n** is an integer equal to or greater than 2. In FIG. **8**, the number **n** of the unit transistors **201** are uniquely indicated by reference characters **201** to **20n**. The unit transistors **201** to **20n** each include the transistor element **61**, the capacitor **62**, and the resistance element **63**.

(74) The transistor element **61**, the capacitor **62**, and the resistance element **63** are almost identical to the transistor element **51**, the capacitor **52**, and the resistance element **53** with respect to electrical characteristics. This means that the unit transistors **101** to **10m** and **201** to **20n** have almost the same electrical characteristics.

(75) The number m , which is the count of the unit transistors **101** included in the amplifier **1401**, and the number n , which is the count of the unit transistors **201** included in the amplifier **1402**, are set as appropriate in accordance with, for example, the output power and output efficiency in the high-power mode and in the low-power mode.

(76) In the present embodiment, the number m and the number n may be, for example, 40 and 20, respectively. In the power amplifier circuit **13**, for example, when only the amplifier **1401** provides amplification in the high-power mode, and only the amplifier **1402** provides amplification in the low-power mode, the ratio of the output power in the high-power mode to the output power in the low-power mode is approximately 40:20, that is, approximately 2:1.

(77) In the power amplifier circuit **13**, when both the amplifiers **1401** and **1402** provide amplification in the high-power mode, and only the amplifier **1402** provides amplification in the low-power mode, the ratio of the output power in the high-power mode to the output power in the low-power mode is approximately 60:20, that is, approximately 3:1.

(78) The first end of the capacitor **52** of each of the unit transistors **101** to **10m** is coupled to the input terminal **31**. The first end of the resistance element **53** of each of the unit transistors **101** to **10m** is coupled to the second end of the impedance circuit **501**, that is, the second end of the capacitor **501a** via the node **N12** and also coupled to the bias circuit **301** via the node **N12**. The collector of the transistor element **51** of each of the unit transistors **101** to **10m** is coupled to the supply-voltage feeding node **N1** and the output terminal **32**.

(79) The first end of the capacitor **62** of each of the unit transistors **201** to **20n** is coupled to the input terminal **31**. The first end of the resistance element **63** of each of the unit transistors **201** to **20n** is coupled to the second end of the impedance circuit **601**, that is, the second end of the capacitor **601a** via the node **N22** and also coupled to the bias circuit **302** via the node **N22**. The collector of the transistor element **61** of each of the unit transistors **201** to **20n** is coupled to the supply-voltage feeding node **N1** and the output terminal **32**.

(80) When the number m of the unit transistors **101** of the amplifier **1401** is greater than the number n of the unit transistors **201** of the amplifier **1402**, the capacitance of the capacitor **501a** is larger than the capacitance of the capacitor **601a**; when the number m of the unit transistors **101** is equal to or smaller than the number n of the unit transistors **201**, the capacitance of the capacitor **501a** is equal to or smaller than the capacitance of the capacitor **601a**.

(81) In the present embodiment, the number m of the unit transistors **101** is greater than the number n of the unit transistors **201**, and thus, the capacitance of the capacitor **501a** is larger than the capacitance of the capacitor **601a**.

(82) The ratio of a capacitance $Ca1$ of the capacitor **501a** to a capacitance $Ca2$ of the capacitor **601a** is in the range of 0.75 to 1.25 times larger than the ratio of the number m of the unit transistors **101** of the amplifier **1401** to the number n of the unit transistors **201** of the amplifier **1402**.

(83) In the present embodiment, $Ca1/Ca2$ is in the range of 0.75 to 1.25 times larger than m/n , that is, 40/20.

(84) With this configuration, the capacitance of the capacitor **501a** corresponds to the number m of the unit transistors **101**, and the capacitance of the capacitor **601a** corresponds to the number n of the unit transistors **201**. As a result, the alternating-current component of the bias current $Ib1$ and the alternating-current component of the bias current $Ib2$ can be set to appropriate values, and thus, it is possible to effectively reduce the gain compression of the transistor elements **51** and **61** and also suppress the degradation of the distortion characteristic of the transistor elements **51** and **61**.

Fourth Embodiment

(85) The following describes a power amplifier circuit according to a fourth embodiment. FIG. 9 is a circuit diagram of a power amplifier circuit **14**. As illustrated in FIG. 9, the power amplifier circuit **14** according to the fourth embodiment differs from the power amplifier circuit **13** according to the third embodiment in that the impedance circuits **501** and **601** each include a resistance

element.

(86) When compared to the impedance circuit **501** of the power amplifier circuit **13** illustrated in FIG. **8**, the impedance circuit **501** of the power amplifier circuit **14** further includes a resistance element **501b** (third resistance element) coupled in series with the capacitor **501a**.

(87) When compared to the impedance circuit **601** of the power amplifier circuit **13** illustrated in FIG. **8**, the impedance circuit **601** of the power amplifier circuit **14** further includes a resistance element **601b** (fourth resistance element) coupled in series with the capacitor **601a**.

(88) In the present embodiment, the capacitor **501a** of the impedance circuit **501** has a first end coupled to the node **N11** and a second end. The resistance element **501b** has a first end coupled to the second end of the capacitor **501a** and a second end coupled to the node **N12**.

(89) Of the impedance circuit **501**, the first end of the capacitor **501a** may be coupled to the node **N12**, and the second end of the resistance element **501b** may be coupled to the node **N11**.

(90) As described above, because the resistance element **501b** is coupled in series with the capacitor **501a** in the impedance circuit **501**, as for the bias current **Ib1**, it is possible to control the alternating-current component of the bias current flowing to the resistance element **53** of each of the unit transistors **101** to **10m**. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **401** of the power amplifier circuit **13** (refer to FIG. **8**).

(91) The capacitor **601a** of the impedance circuit **601** has a first end coupled to the node **N21** and a second end. The resistance element **601b** has a first end coupled to the second end of the capacitor **601a** and a second end coupled to the node **N22**.

(92) Of the impedance circuit **601**, the first end of the capacitor **601a** may be coupled to the node **N22**, and the second end of the resistance element **601b** may be coupled to the node **N21**.

(93) As described above, because the resistance element **601b** is coupled in series with the capacitor **601a** in the impedance circuit **601**, as for the bias current **Ib2**, it is possible to control the alternating-current component of the bias current flowing to the resistance element **63** of each of the unit transistors **201** to **20n**. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **402** of the power amplifier circuit **13** (refer to FIG. **8**).

Fifth Embodiment

(94) The following describes a power amplifier circuit according to a fifth embodiment. FIG. **10** is a circuit diagram of a power amplifier circuit **15**. As illustrated in FIG. **10**, the power amplifier circuit **15** according to the fifth embodiment differs from the power amplifier circuit **13** according to the third embodiment in that the impedance circuits **501** and **601** each include an inductor.

(95) When compared to the impedance circuit **501** of the power amplifier circuit **13** illustrated in FIG. **8**, the impedance circuit **501** of the power amplifier circuit **15** further includes an inductor **501c** (first inductor) coupled in series with the capacitor **501a**.

(96) When compared to the impedance circuit **601** of the power amplifier circuit **13** illustrated in FIG. **8**, the impedance circuit **601** of the power amplifier circuit **15** further includes an inductor **601c** (second inductor) coupled in series with the capacitor **601a**.

(97) In the present embodiment, the capacitor **501a** of the impedance circuit **501** has the first end coupled to the node **N11** and the second end. The inductor **501c** has a first end coupled to the second end of the capacitor **501a** and a second end coupled to the node **N12**.

(98) Of the impedance circuit **501**, the first end of the capacitor **501a** may be coupled to the node **N12**, and the second end of the inductor **501c** may be coupled to the node **N11**.

(99) As described above, because the inductor **501c** is coupled in series with the capacitor **501a** in the impedance circuit **501**, as for the bias current **Ib1**, it is possible to control the alternating-current component of the bias current flowing to the resistance element **53** of each of the unit transistors **101** to **10m**. Specifically, for example, by appropriately controlling the resonant frequency of a series resonant circuit constituted by the capacitor **501a** and the inductor **501c** coupled in series with each other, it is possible to control the alternating-current component of the bias current in response to the frequency of the alternating-current component. As a result, it is possible to more

highly accurately reduce the gain compression than the amplifier **401** of the power amplifier circuit **13** (refer to FIG. **8**).

(100) The capacitor **601a** of the impedance circuit **601** has the first end coupled to the node **N21** and the second end. The inductor **601c** has a first end coupled to the second end of the capacitor **601a** and a second end coupled to the node **N22**.

(101) Of the impedance circuit **601**, the first end of the capacitor **601a** may be coupled to the node **N22**, and the second end of the inductor **601c** may be coupled to the node **N21**.

(102) As described above, because the inductor **601c** is coupled in series with the capacitor **601a** in the impedance circuit **601**, as for the bias current **Ib2**, it is possible to control the alternating-current component of the bias current flowing to the resistance element **63** of each of the unit transistors **201** to **20n**. Specifically, for example, by appropriately controlling the resonant frequency of a series resonant circuit constituted by the capacitor **601a** and the inductor **601c** coupled in series with each other, it is possible to control the alternating-current component of the bias current in response to the frequency of the alternating-current component. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **402** of the power amplifier circuit **13** (refer to FIG. **8**).

Sixth Embodiment

(103) The following describes a power amplifier circuit according to a sixth embodiment. FIG. **11** is a circuit diagram of a power amplifier circuit **16**. As illustrated in FIG. **11**, the power amplifier circuit **16** according to the sixth embodiment differs from the power amplifier circuit **13** according to the third embodiment in that a matching circuit is provided in a stage before the amplifiers.

(104) When compared to the power amplifier circuit **13** illustrated in FIG. **8**, the power amplifier circuit **16** further includes a matching circuit **701** (first matching circuit).

(105) The matching circuit **701** has a first end coupled to the input terminal **31**, the first end of the impedance circuit **501**, and the first end of the impedance circuit **601** and a second end coupled to the base of the transistor element **51** and the base of the transistor element **61**.

(106) In the present embodiment, the matching circuit **701** includes a capacitor **701a** and an inductor **701b**. The capacitor **701a** has a first end coupled to the input terminal **31**, the first end of the capacitor **501a**, and the first end of the capacitor **601a**. The capacitor **701a** also has a second end coupled to the first end of the capacitor **52** of each of the unit transistors **101** to **10m** and the first end of the capacitor **62** of each of the unit transistors **201** to **20n**.

(107) The inductor **701b** has a first end coupled to the first end of the capacitor **701a** and a second end coupled to the ground.

(108) For example, because the impedance circuits **501** and **601** are provided, an impedance mismatch may be caused between a circuit in a stage before the power amplifier circuit **16** and the amplifiers **1401** and **1402**. With consideration of this problem, the matching circuit **701** is provided between the input terminal **31**, the first end of the impedance circuit **501**, and the first end of the impedance circuit **601**, and the amplifiers **1401** and **1402**, so that it is possible to improve the impedance mismatch due to the impedance circuits **501** and **601**. This configuration can effectively match the impedance between the circuit in a stage before the power amplifier circuit **16** and the amplifiers **1401** and **1402**.

Seventh Embodiment

(109) The following describes a power amplifier circuit according to a seventh embodiment. FIG.

12 is a circuit diagram of a power amplifier circuit **17**. As illustrated in FIG. **12**, the power amplifier circuit **17** according to the seventh embodiment differs from the power amplifier circuit **16** according to the sixth embodiment in that the matching circuit is provided between the input terminal **31**, and the first end of the impedance circuit **501** and the first end of the impedance circuit **601**.

(110) In the present embodiment, the matching circuit **701** has a first end coupled to the input terminal **31** and a second end coupled to the first end of the impedance circuit **501**, the first end of

the impedance circuit **601**, the base of the transistor element **51**, and the base of the transistor element **61**.

(111) Specifically, the capacitor **701a** of the matching circuit **701** has a first end coupled to the input terminal **31** and a second end coupled to the first end of the capacitor **501a**, the first end of the capacitor **601a**, the first end of the capacitor **52** of each of the unit transistors **101** to **10m**, and the first end of the capacitor **62** of each of the unit transistors **201** to **20n**.

(112) The inductor **701b** has the first end coupled to the first end of the capacitor **701a** and the second end coupled to the ground.

(113) For example, because the impedance circuits **501** and **601** are provided, an impedance mismatch may be caused between a circuit in a stage before the power amplifier circuit **17** and the amplifiers **1401** and **1402**. With consideration of this problem, the matching circuit **701** is provided between the input terminal **31**, and the first end of the impedance circuit **501**, the first end of the impedance circuit **601**, and the amplifiers **1401** and **1402**, so that it is possible to improve the impedance mismatch due to the impedance circuits **501** and **601**. This configuration can effectively match the impedance between the circuit in a stage before the power amplifier circuit **17** and the amplifiers **1401** and **1402**.

(114) The capacitance (hereinafter also referred to as second capacitance) of the capacitor **501a** is larger than when the impedance circuit **501** is coupled between the input terminal **31** and the matching circuit **701** (hereinafter, the capacitance of the capacitor **501a** in this case is also referred to as first capacitance).

(115) In other words, the first capacitance, which is the capacitance of the capacitor **501a** of the power amplifier circuit **16** illustrated in FIG. **11**, is smaller than the second capacitance, which is the capacitance of the capacitor **501a** obtained when the impedance circuit **501** is coupled between the matching circuit **701**, and the base of the transistor element **51** and the base of the transistor element **61**.

(116) Here, the first capacitance is the capacitance of the capacitor **501a** obtained when the power amplifier circuit **16** illustrated in FIG. **11** can effectively achieve impedance matching between the circuit in a stage before the power amplifier circuit **16**, and the amplifiers **1401** and **1402** and also effectively reduce the gain compression of the amplifier **1401**.

(117) The second capacitance is the capacitance of the capacitor **501a** obtained when the power amplifier circuit **17** illustrated in FIG. **12** can effectively achieve impedance matching between the circuit in a stage before the power amplifier circuit **17**, and the amplifiers **1401** and **1402** and also effectively reduce the gain compression of the amplifier **1401**.

(118) The capacitance (hereinafter also referred to as fourth capacitance) of the capacitor **601a** is larger than when the impedance circuit **601** is coupled between the input terminal **31** and the matching circuit **701** (hereinafter, the capacitance of the capacitor **601a** in this case is also referred to as third capacitance).

(119) In other words, the third capacitance, which is the capacitance of the capacitor **601a** of the power amplifier circuit **16** illustrated in FIG. **11**, is smaller than the fourth capacitance, which is the capacitance of the capacitor **601a** obtained when the impedance circuit **601** is coupled between the matching circuit **701**, and the base of the transistor element **51** and the base of the transistor element **61**.

(120) Here, the third capacitance is the capacitance of the capacitor **601a** obtained when the power amplifier circuit **16** illustrated in FIG. **11** can effectively achieve impedance matching between the circuit in a stage before the power amplifier circuit **16**, and the amplifiers **1401** and **1402** and also effectively reduce the gain compression of the amplifier **1402**.

(121) The fourth capacitance is the capacitance of the capacitor **601a** obtained when the power amplifier circuit **17** illustrated in FIG. **12** can effectively achieve impedance matching between the circuit in a stage before the power amplifier circuit **17**, and the amplifiers **1401** and **1402** and also effectively reduce the gain compression of the amplifier **1402**.

(122) Specifically, the second capacitance of the capacitor **501a** and the fourth capacitance of the capacitor **601a** illustrated in FIG. **12** are respectively larger than the first capacitance of the capacitor **501a** and the third capacitance of the capacitor **601a** illustrated in FIG. **11**.

(123) As such, it is possible to achieve suitable values of the first capacitance and the second capacitance with consideration of the capacitance of the capacitor **701a** of the matching circuit **701**, and as a result, the alternating-current component of the bias current **Ib1** can be caused to adequately flow to the base of the transistor element **51** through the impedance circuit **501**. As a result, it is possible to adequately reduce the gain compression of the transistor element **51** and also suppress the degradation of the distortion characteristic of the transistor element **51**.

(124) Further, it is possible to achieve suitable values of the third capacitance and the fourth capacitance with consideration of the capacitance of the capacitor **701a** of the matching circuit **701**, and as a result, the alternating-current component of the bias current **Ib2** can be caused to adequately flow to the base of the transistor element **61** through the impedance circuit **601**. As a result, it is possible to adequately reduce the gain compression of the transistor element **61** and also suppress the degradation of the distortion characteristic of the transistor element **61**.

Eighth Embodiment

(125) The following describes a power amplifier circuit according to an eighth embodiment. FIG. **13** is a circuit diagram of a power amplifier circuit **18**. As illustrated in FIG. **13**, the power amplifier circuit **18** according to the eighth embodiment differs from the power amplifier circuit **16** according to the sixth embodiment in that the power amplifier circuit **18** includes amplifiers in two stages.

(126) When compared to the power amplifier circuit **16** illustrated in FIG. **11**, the power amplifier circuit **18** further includes a bias circuit **303**, an amplifier **403**, and a matching circuit **702** (second matching circuit).

(127) The matching circuit **702** is configured in the same manner as, for example, the matching circuit **701**. The bias circuit **303** is configured in the same manner as, for example, the bias circuit **301** or **302**.

(128) The amplifier **403** includes a unit transistor **151**. The amplifier **403** functions as a first-stage amplifier. The unit transistor **151** includes a transistor element **71** (third transistor), a capacitor **72** (impedance circuit), and a resistance element **73**.

(129) The capacitor **72** of the amplifier **403** has a first end coupled to the input terminal **31** via the matching circuit **702** and a second end. The transistor element **71** has a base coupled to the second end of the capacitor **72**, a collector coupled to a supply-voltage feeding node **N4** for feeding a supply voltage **VCC2** and also coupled to the amplifiers **1401** and **1402** via the matching circuit **701**, and an emitter coupled to the ground. Specifically, the collector of the transistor element **71** is coupled to the base of the transistor element **51** via the matching circuit **701** and the capacitor **52** and also to the base of the transistor element **61** via the matching circuit **701** and the capacitor **62**.

(130) The resistance element **73** has a first end coupled to the bias circuit **303** and a second end coupled to the base of the transistor element **71**. The bias circuit **303** supplies a bias to the base of the transistor element **71** through the resistance element **73**.

(131) The impedance circuit **501** has a first end coupled between the base of the transistor element **71** and the matching circuit **702** and a second end coupled between the bias circuit **301** and the resistance element **53**.

(132) Specifically, the capacitor **501a** of the impedance circuit **501** has a first end coupled to a node **N13** provided in a wire **W3** connecting the matching circuit **702** and the first end of the capacitor **72** and a second end coupled to the node **N12**.

(133) The impedance circuit **601** has a first end coupled between the base of the transistor element **71** and the matching circuit **702** and a second end coupled between the bias circuit **302** and the resistance element **63**.

(134) Specifically, the capacitor **601a** of the impedance circuit **601** has a first end coupled to a node

N23 provided in the wire W3 and a second end coupled to the node N22.

(135) As described above, the impedance circuits **501** and **601** are not coupled between the first-stage amplifier **403** and the final-stage amplifiers **1401** and **1402**, and as a result, it is possible to reduce the likelihood of a mismatch of the input of the final-stage amplifiers **1401** and **1402**.

Consequently, in the power amplifier circuit **18** including amplifiers in two stages, it is possible to suppress the degradation of the distortion characteristic of the final-stage amplifiers **1401** and **1402** due to the impedance circuits **501** and **601**.

Ninth Embodiment

(136) The following describes a power amplifier circuit according to a ninth embodiment. FIG. **14** is a circuit diagram of a power amplifier circuit **19**. As illustrated in FIG. **14**, the power amplifier circuit **19** according to the ninth embodiment differs from the power amplifier circuit **17** according to the seventh embodiment in that the power amplifier circuit **19** includes amplifiers in two stages.

(137) When compared to the power amplifier circuit **17** illustrated in FIG. **12**, the power amplifier circuit **19** further includes the bias circuit **303**, the amplifier **403**, and the matching circuit **702**.

(138) The bias circuit **303**, the amplifier **403**, and the matching circuit **702** of the power amplifier circuit **19** are configured in the same manner as the bias circuit **303**, the amplifier **403**, and the matching circuit **702** of the power amplifier circuit **18** illustrated in FIG. **13**.

(139) The collector of the transistor element **71** of the amplifier **403** is coupled to the supply-voltage feeding node N4 and also to the first end of the capacitor **52** and the first end of the capacitor **62** via the matching circuit **701**.

(140) The first end of the capacitor **501a** of the impedance circuit **501** is coupled to the collector of the transistor element **71** via the capacitor **701a** of the matching circuit **701**. The first end of the capacitor **601a** of the impedance circuit **601** is coupled to the collector of the transistor element **71** via the capacitor **701a** of the matching circuit **701**.

(141) For example, because the impedance circuits **501** and **601** are provided, an impedance mismatch may be caused between the amplifier **403** and the amplifiers **1401** and **1402**. With consideration of this problem, the matching circuit **701** is provided between the amplifier **403**, and the first end of the impedance circuit **501**, the first end of the impedance circuit **601**, and the amplifiers **1401** and **1402**, so that it is possible to improve the impedance mismatch due to the impedance circuits **501** and **601**. This configuration can effectively match the impedance between the amplifier **403** and the amplifiers **1401** and **1402**.

(142) In the above description, the first end of the capacitor **501a** of the impedance circuit **501** is coupled to the collector of the transistor element **71** via the capacitor **701a** of the matching circuit **701**, but this should not be construed in a limiting sense. The first end of the capacitor **501a** may be coupled directly to the collector of the transistor element **71**. This configuration can also effectively match the impedance between the amplifier **403** and the amplifiers **1401** and **1402**.

(143) Further, in the above description, the first end of the capacitor **601a** of the impedance circuit **601** is coupled to the collector of the transistor element **71** via the capacitor **701a** of the matching circuit **701**, but this should not be construed in a limiting sense. The first end of the capacitor **601a** may be coupled directly to the collector of the transistor element **71**. This configuration can also effectively match the impedance between the amplifier **403** and the amplifiers **1401** and **1402**.

Tenth Embodiment

(144) The following describes a power amplifier circuit according to a tenth embodiment. FIG. **15** is a circuit diagram of a power amplifier circuit **20**. As illustrated in FIG. **15**, the power amplifier circuit **20** according to the tenth embodiment differs from the power amplifier circuit **19** according to the ninth embodiment in that the power amplifier circuit **20** includes amplifiers in three stages.

(145) When compared to the power amplifier circuit **19** illustrated in FIG. **14**, the power amplifier circuit **20** further includes a bias circuit **304**, an amplifier **404**, and a matching circuit **703**.

(146) The matching circuit **703** is configured in the same manner as, for example, the matching circuit **701**. The bias circuit **304** is configured in the same manner as, for example, the bias circuit

301 or 302.

(147) The amplifier **404** includes a unit transistor **161**. The amplifier **404** functions as a first-stage amplifier. The unit transistor **161** includes a transistor element **81**, a capacitor **82** (impedance circuit), and a resistance element **83**.

(148) The capacitor **82** of the amplifier **404** has a first end coupled to the input terminal **31** via the matching circuit **703** and a second end. The transistor element **81** has a base coupled to the second end of the capacitor **82**, a collector coupled to a supply-voltage feeding node **N5** for feeding a supply voltage **VCC3** and also coupled to the amplifier **403** via the matching circuit **702**, and an emitter coupled to the ground. Specifically, the collector of the transistor element **81** is coupled to the base of the transistor element **71** via the matching circuit **702** and the capacitor **72**.

(149) The resistance element **83** has a first end coupled to the bias circuit **304** and a second end coupled to the base of the transistor element **81**. The bias circuit **304** supplies a bias to the base of the transistor element **81** through the resistance element **83**. The amplifier **403** functions as a second-stage amplifier.

(150) In the above description of the power amplifier circuits **11** to **20**, the transistor elements **51** and **61** respectively correspond to the bias circuits **301** and **302** so that the output level is switched between two levels of the high-power mode and the low-power mode, but this should not be construed in a limiting sense. Three or more bias circuits may be provided so that the output level is switched among three or more levels.

(151) Furthermore, in the above description of the power amplifier circuits **13** to **20**, the number *m* of the unit transistors **101** is two or more, but this should not be construed in a limiting sense. The number *m* of the unit transistors **101** may be one.

(152) Moreover, in the above description of the power amplifier circuits **13** to **20**, the number *n* of the unit transistors **201** is two or more, but this should not be construed in a limiting sense. The number *n* of the unit transistors **201** may be one.

(153) Further, in the above description of the power amplifier circuits **11** to **20**, the transistors **301a**, **301b**, and **301c** of the bias circuits **301** and **302** are bipolar transistors, but this should not be construed in a limiting sense. The transistors **301a**, **301b**, and **301c** of the bias circuits **301** and **302** may be field effect transistors.

(154) Furthermore, in the description of the power amplifier circuits **11** to **20**, the unit transistor **101** includes the capacitor **52**, but this should not be construed in a limiting sense. The unit transistor **101** does not necessarily include the capacitor **52**. A resistance element or an inductor may be coupled in series with the capacitor **52**.

(155) Moreover, in the description of the power amplifier circuits **11** to **20**, the unit transistor **201** includes the capacitor **62**, but this should not be construed in a limiting sense. The unit transistor **201** does not necessarily include the capacitor **62**. A resistance element or an inductor may be coupled in series with the capacitor **62**.

(156) The exemplary embodiments of the present disclosure have been described above. The power amplifier circuits **11** to **20** include the transistor element **51**, the resistance element **53**, the bias circuit **301**, the transistor element **61**, the resistance element **63**, the bias circuit **302**, and the impedance circuit **501**. The transistor element **51** has the base coupled to the input terminal **31**, the collector coupled to the output terminal **32**, and the emitter coupled to the ground. The bias circuit **301** is coupled to the base of the transistor element **51** via the resistance element **53**. The bias circuit **301** supplies a bias to the transistor element **51**. The transistor element **61** has the base coupled to the input terminal **31**, the collector coupled to the output terminal **32**, and the emitter coupled to the ground. The bias circuit **302** is coupled to the base of the transistor element **61** via the resistance element **63**. The bias circuit **302** supplies a bias to the transistor element **61**. The impedance circuit **501** has the first end coupled between the base of the transistor element **51** and the input terminal **31** and the second end coupled between the bias circuit **301** and the resistance element **53**. The impedance circuit **501** is opened for direct-current components and closed for

alternating-current components.

(157) As such, the bias circuits **301** and **302** respectively supply a bias to the transistor elements **51** and **61**. As a result, for example, the transistor elements **51** and **61** are caused to provide amplification in the high-power mode, whereas only the transistor element **61** is caused to provide amplification in the low-power mode. Accordingly, the output level of the power amplifier circuits **11** to **20** can be changed between two levels. Because the impedance circuit **501** is provided in parallel with the resistance element **53**, the alternating-current component of the bias current I_{b1} flowing from the bias circuit **301** to the base of the transistor element **51** can be caused to flow through the impedance circuit **501** to the base of the transistor element **51**. As a result, without lowering the resistance of the resistance element **53**, this can achieve the same effect for alternating current as lowering the resistance of the resistance element **53**, and consequently, it is possible to reduce the gain compression. By reducing the gain compression of the transistor element **51**, the linearity of the amplification characteristic can be maintained. Consequently, for example, it is possible to improve the distortion characteristic of the power amplifier circuits **11** to **20** in the high-power mode.

(158) The power amplifier circuits **12** to **20** further include the impedance circuit **601**. The impedance circuit **601** has the first end coupled between the input terminal **31** and the base of the transistor element **61** and the second end coupled between the bias circuit **302** and the resistance element **63**. The impedance circuit **601** is opened for direct-current components and closed for alternating-current components.

(159) Because the impedance circuit **601** is provided in parallel with the resistance element **63**, the alternating-current component of the bias current I_{b2} flowing from the bias circuit **302** to the base of the transistor element **61** can be caused to flow through the impedance circuit **601** to the base of the transistor element **61**. As a result, without lowering the resistance of the resistance element **63**, this can achieve the same effect for alternating current as lowering the resistance of the resistance element **63**, and consequently, it is possible to reduce the gain compression of the transistor element **61**. By reducing the gain compression of the transistor element **61**, the linearity of the amplification characteristic of the transistor element **61** can be maintained. Consequently, for example, it is possible to improve the distortion characteristic of the power amplifier circuits **12** to **20** in the low-power mode.

(160) Because the bias circuits **301** and **302** are separated from each other with respect to direct-current components, the bias circuits **301** and **302** can adequately supply a bias of a direct-current component respectively to the base of the transistor element **51** and the base of the transistor element **61**. Because the impedance circuits **501** and **601** are respectively provided in parallel with the resistance elements **53** and **63**, the alternating-current component of the bias current I_{b1} and the alternating-current component of the bias current I_{b2} are caused to respectively flow to the impedance circuits **501** and **601**. The transistor elements **51** and **61** can thus provide amplification with balance. Further, because voltage drop across the resistance elements **53** and **63** is reduced, it is possible to suppress the gain compression of the transistor elements **51** and **61** regardless of the operation mode. This means that the distortion characteristic of the power amplifier circuits **11** to **20** can be improved regardless of the operation mode.

(161) The power amplifier circuits **12** to **20** include the number m of the unit transistors **101** each having the resistance element **53** and the transistor element **51** and also include the number n of the unit transistors **201** each having the resistance element **63** and the transistor element **61**. The impedance circuit **501** includes the capacitor **501a**. The impedance circuit **601** includes the capacitor **601a**. When the number m of the unit transistors **101** is greater than the number n of the unit transistors **201**, the capacitance of the capacitor **501a** is larger than the capacitance of the capacitor **601a**; when the number m of the unit transistors **101** is equal to or smaller than the number n of the unit transistors **201**, the capacitance of the capacitor **501a** is equal to or smaller than the capacitance of the capacitor **601a**.

(162) With this configuration, the capacitance of the capacitor **501a** corresponds to the number m of the unit transistors **101**, and the capacitance of the capacitor **601a** corresponds to the number n of the unit transistors **201**. As a result, the alternating-current component of the bias current I_{b1} and the alternating-current component of the bias current I_{b2} can be set to appropriate values, and thus, it is possible to effectively reduce the gain compression of the transistor elements **51** and **61** and also suppress the degradation of the distortion characteristic of the transistor elements **51** and **61**.

(163) The power amplifier circuits **12** to **20** include the number m of the unit transistors **101** each having the resistance element **53** and the transistor element **51** and also include the number n of the unit transistors **201** each having the resistance element **63** and the transistor element **61**. The impedance circuit **501** includes the capacitor **501a**. The impedance circuit **601** includes the capacitor **601a**. The ratio of the capacitance of the capacitor **501a** to the capacitance of the capacitor **601a** is in the range of 0.75 to 1.25 times larger than the ratio of the number m of the unit transistors **101** to the number n of the unit transistors **201**.

(164) With this configuration, the capacitance of the capacitor **501a** corresponds to the number m of the unit transistors **101**, and the capacitance of the capacitor **601a** corresponds to the number n of the unit transistors **201**. As a result, the alternating-current component of the bias current I_{b1} and the alternating-current component of the bias current I_{b2} can be individually set to appropriate values, and thus, it is possible to effectively reduce the gain compression of the transistor elements **51** and **61** and also suppress the degradation of the distortion characteristic of the transistor elements **51** and **61**.

(165) The power amplifier circuit **16** further includes the matching circuit **701**. The matching circuit **701** has the first end coupled to the input terminal **31**, the first end of the impedance circuit **501**, and the first end of the impedance circuit **601** and the second end coupled to the base of the transistor element **51** and the base of the transistor element **61**.

(166) For example, because the impedance circuits **501** and **601** are provided, an impedance mismatch may be caused between a circuit in a stage before the power amplifier circuit **16**, and the base of the transistor element **51** and the base of the transistor element **61**. With consideration of this problem, the matching circuit **701** is provided between the input terminal **31**, the first end of the impedance circuit **501**, and the first end of the impedance circuit **601**, and the base of the transistor element **51** and the base of the transistor element **61**, so that it is possible to improve the impedance mismatch due to the impedance circuits **501** and **601**. As a result, while suppressing the degradation of the distortion characteristic of the transistor elements **51** and **61** by using the impedance circuits **501** and **601**, it is possible to effectively match the impedance between the circuit in a stage before the power amplifier circuit **16**, and the base of the transistor element **51** and the base of the transistor element **61**.

(167) In the power amplifier circuit **16**, the impedance circuit **501** includes the capacitor **501a**. The impedance circuit **601** includes the capacitor **601a**. The first capacitance, which is the capacitance of the capacitor **501a**, is smaller than the second capacitance, which is the capacitance of the capacitor **501a** obtained when the impedance circuit **501** is coupled between the matching circuit **701**, and the base of the transistor element **51** and the base of the transistor element **61**. The third capacitance, which is the capacitance of the capacitor **601a**, is smaller than the fourth capacitance, which is the capacitance of the capacitor **601a** obtained when the impedance circuit **601** is coupled between the matching circuit **701**, and the base of the transistor element **51** and the base of the transistor element **61**.

(168) As such, it is possible to achieve a suitable value of the first capacitance with consideration of, for example, the capacitance of the matching circuit **701**, and as a result, the alternating-current component of the bias current I_{b1} can be caused to adequately flow to the base of the transistor element **51** through the impedance circuit **501**. As a result, it is possible to adequately reduce the gain compression of the transistor element **51** and also suppress the degradation of the distortion characteristic of the transistor element **51**. Further, it is possible to achieve a suitable value of the

third capacitance with consideration of, for example, the capacitance of the matching circuit **701**, and as a result, the alternating-current component of the bias current **Ib2** can be caused to adequately flow to the base of the transistor element **61** through the impedance circuit **601**. As a result, it is possible to adequately reduce the gain compression of the transistor element **61** and also suppress the degradation of the distortion characteristic of the transistor element **61**.

(169) The power amplifier circuit **17** further includes the matching circuit **701**. The matching circuit **701** has the first end coupled to the input terminal **31** and the second end coupled to the first end of the impedance circuit **501**, the first end of the impedance circuit **601**, the base of the transistor element **51**, and the base of the transistor element **61**.

(170) For example, because the impedance circuits **501** and **601** are provided, an impedance mismatch may be caused between a circuit in a stage before the power amplifier circuit **17**, and the base of the transistor element **51** and the base of the transistor element **61**. With consideration of this problem, the matching circuit **701** is provided between the input terminal **31**, and the first end of the impedance circuit **501**, the first end of the impedance circuit **601**, and the base of the transistor element **51** and the base of the transistor element **61**, so that it is possible to improve the impedance mismatch due to the impedance circuits **501** and **601**. As a result, while suppressing the degradation of the distortion characteristic of the transistor elements **51** and **61** by using the impedance circuits **501** and **601**, it is possible to effectively match the impedance between the circuit in a stage before the power amplifier circuit **17**, and the base of the transistor element **51** and the base of the transistor element **61**.

(171) In the power amplifier circuit **17**, the impedance circuit **501** includes the capacitor **501a**. The impedance circuit **601** includes the capacitor **601a**. The second capacitance, which is the capacitance of the capacitor **501a**, is larger than the first capacitance, which is the capacitance of the capacitor **501a** obtained when the impedance circuit **501** is coupled between the input terminal **31** and the matching circuit **701**. The fourth capacitance, which is the capacitance of the capacitor **601a**, is larger than the third capacitance, which is the capacitance of the capacitor **601a** obtained when the impedance circuit **601** is coupled between the input terminal **31** and the matching circuit **701**.

(172) As such, it is possible to achieve a suitable value of the second capacitance with consideration of, for example, the capacitance of the matching circuit **701**, and as a result, the alternating-current component of the bias current **Ib1** can be caused to adequately flow to the base of the transistor element **51** through the impedance circuit **501**. As a result, it is possible to adequately reduce the gain compression of the transistor element **51** and also suppress the degradation of the distortion characteristic of the transistor element **51**. Further, it is possible to achieve a suitable value of the fourth capacitance with consideration of, for example, the capacitance of the matching circuit **701**, and as a result, the alternating-current component of the bias current **Ib2** can be caused to adequately flow to the base of the transistor element **61** through the impedance circuit **601**. As a result, it is possible to adequately reduce the gain compression of the transistor element **61** and also suppress the degradation of the distortion characteristic of the transistor element **51**.

(173) In the power amplifier circuits **12** to **20**, the impedance circuit **501** includes the capacitor **501a**.

(174) With this configuration, the capacitor **501a** can easily implement the impedance circuit **501** configured to be opened for direct-current components and closed for alternating-current components.

(175) In the power amplifier circuits **12** to **20**, the impedance circuit **601** includes the capacitor **601a**.

(176) With this configuration, the capacitor **601a** can easily implement the impedance circuit **601** configured to be opened for direct-current components and closed for alternating-current components.

(177) In the power amplifier circuit **14**, the impedance circuit **501** further includes the resistance element **501b** coupled in series with the capacitor **501a**.

(178) As described above, because the resistance element **501b** is coupled in series with the capacitor **501a** in the impedance circuit **501**, as for the bias current I_{b1} , it is possible to control the alternating-current component of the bias current flowing to the resistance element **53** of each of the unit transistors **101** to **10m**. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **401** of the power amplifier circuit **13** (refer to FIG. **8**).

(179) In the power amplifier circuit **15**, the impedance circuit **501** further includes the inductor **501c** coupled in series with the capacitor **501a**.

(180) As described above, because the inductor **501c** is coupled in series with the capacitor **501a** in the impedance circuit **501**, as for the bias current I_{b1} , it is possible to control the alternating-current component of the bias current flowing to the resistance element **53** of each of the unit transistors **101** to **10m**. Specifically, for example, by appropriately controlling the resonant frequency of a series resonant circuit constituted by the capacitor **501a** and the inductor **501c** coupled in series with each other, it is possible to control the alternating-current component of the bias current in response to the frequency of the alternating-current component. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **401** of the power amplifier circuit **13** (refer to FIG. **8**).

(181) In the power amplifier circuit **14**, the impedance circuit **601** further includes the resistance element **601b** coupled in series with the capacitor **601a**.

(182) As described above, because the resistance element **601b** is coupled in series with the capacitor **601a** in the impedance circuit **601**, as for the bias current I_{b2} , it is possible to control the alternating-current component of the bias current flowing to the resistance element **63** of each of the unit transistors **201** to **20n**. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **402** of the power amplifier circuit **13** (refer to FIG. **8**).

(183) In the power amplifier circuit **15**, the impedance circuit **601** further includes the inductor **601c** coupled in series with the capacitor **601a**.

(184) As described above, because the inductor **601c** is coupled in series with the capacitor **601a** in the impedance circuit **601**, as for the bias current I_{b2} , it is possible to control the alternating-current component of the bias current flowing to the resistance element **63** of each of the unit transistors **201** to **20n**. Specifically, for example, by appropriately controlling the resonant frequency of a series resonant circuit constituted by the capacitor **601a** and the inductor **601c** coupled in series with each other, it is possible to control the alternating-current component of the bias current in response to the frequency of the alternating-current component. As a result, it is possible to more highly accurately reduce the gain compression than the amplifier **402** of the power amplifier circuit **13** (refer to FIG. **8**).

(185) In the power amplifier circuits **11** to **20**, the first end of the impedance circuit **501** is coupled to the node **N11** provided in the wire connecting the input terminal **31** and the base of the transistor element **51**. The length **L11** of the wire from the input terminal **31** to the node **N11** is more than 0.5 times longer than the length **L10** of the wire from the input terminal **31** to the base of the transistor element **51**.

(186) With this configuration, the alternating-current component of the bias current I_{b1} flowing from the bias circuit **301** to the base of the transistor element **51** can be caused to efficiently flow through the impedance circuit **501** to the base of the transistor element **51**. As a result, it is possible to effectively reduce the gain compression of the transistor element **51** and also suppress the degradation of the distortion characteristic of the transistor element **51**.

(187) In the power amplifier circuits **11** to **20**, the first end of the impedance circuit **601** is coupled to the node **N21** provided in the wire connecting the input terminal **31** and the base of the transistor element **61**. The length **L21** of the wire from the input terminal **31** to the node **N21** is more than 0.5 times longer than the length of the wire from the input terminal **31** to the base of the transistor

element **61**.

(188) With this configuration, the alternating-current component of the bias current I_{b2} flowing from the bias circuit **302** to the base of the transistor element **61** can be caused to efficiently flow through the impedance circuit **601** to the base of the transistor element **61**. As a result, it is possible to effectively reduce the gain compression of the transistor element **61** and also suppress the degradation of the distortion characteristic of the transistor element **61**.

(189) The power amplifier circuit **18** includes the transistor element **51**, the resistance element **53**, the bias circuit **301**, the transistor element **61**, the resistance element **63**, the bias circuit **302**, the matching circuit **701** and **702**, the transistor element **71**, and the impedance circuits **501** and **601**. The transistor element **51** has the base, the collector coupled to the output terminal **32**, and the emitter coupled to the ground. The bias circuit **301** is coupled to the base of the transistor element **51** via the resistance element **53**. The bias circuit **301** supplies a bias to the transistor element **51**. The transistor element **61** has the base, the collector coupled to the output terminal **32**, and the emitter coupled to the ground. The bias circuit **302** is coupled to the base of the transistor element **61** via the resistance element **63**. The bias circuit **302** supplies a bias to the transistor element **61**. The transistor element **71** has the base coupled to the input terminal **31** via the matching circuit **702**, the collector coupled to the base of the transistor element **51** and the base of the transistor element **61** via the matching circuit **701**, and the emitter coupled to the ground. The impedance circuit **501** has the first end coupled between the base of the transistor element **71** and the matching circuit **702** and the second end coupled between the bias circuit **301** and the resistance element **53**. The impedance circuit **601** is opened for direct-current components and closed for alternating-current components. The impedance circuit **601** has the first end coupled between the base of the transistor element **71** and the matching circuit **702** and the second end coupled between the bias circuit **302** and the resistance element **63**. The impedance circuit **601** is opened for direct-current components and closed for alternating-current components.

(190) As such, the bias circuits **301** and **302** respectively supply a bias to the transistor elements **51** and **61**. As a result, for example, the transistor elements **51** and **61** are caused to provide amplification in the high-power mode, whereas only the transistor element **61** is caused to provide amplification in the low-power mode. Accordingly, the output level of the power amplifier circuit **18** can be changed between two levels. Because the impedance circuit **501** is provided in parallel with the resistance element **53**, the alternating-current component of the bias current I_{b1} flowing from the bias circuit **301** to the base of the transistor element **51** can be caused to flow through the impedance circuit **501**, the transistor element **71**, and the matching circuit **701** to the base of the transistor element **51**. As a result, without lowering the resistance of the resistance element **53**, this can achieve the same effect for alternating current as lowering the resistance of the resistance element **53**, and consequently, it is possible to reduce the gain compression. By reducing the gain compression of the transistor element **51**, the linearity of the amplification characteristic can be maintained. Consequently, for example, it is possible to improve the distortion characteristic of the power amplifier circuit **18** in the high-power mode.

(191) Because the impedance circuit **601** is provided in parallel with the resistance element **63**, the alternating-current component of the bias current I_{b2} flowing from the bias circuit **302** to the base of the transistor element **61** can be caused to flow through the impedance circuit **601**, the transistor element **71**, and the matching circuit **701** to the base of the transistor element **61**. As a result, without lowering the resistance of the resistance element **63**, this can achieve the same effect for alternating current as lowering the resistance of the resistance element **63**, and consequently, it is possible to reduce the gain compression of the transistor element **61**. By reducing the gain compression of the transistor element **61**, the linearity of the amplification characteristic of the transistor element **61** can be maintained. Consequently, for example, it is possible to improve the distortion characteristic of the power amplifier circuit **18** in the low-power mode.

(192) Because the bias circuits **301** and **302** are separated from each other with respect to direct-

current components, the bias circuits **301** and **302** can adequately supply a bias of a direct-current component respectively to the base of the transistor element **51** and the base of the transistor element **61**. Because the impedance circuits **501** and **601** are respectively provided in parallel with the resistance elements **53** and **63**, the alternating-current component of the bias current **Ib1** and the alternating-current component of the bias current **Ib2** are caused to respectively flow to the impedance circuits **501** and **601**. The transistor elements **51** and **61** can thus provide amplification with balance. Further, because voltage drop across the resistance elements **53** and **63** is reduced, it is possible to suppress the gain compression of the transistor elements **51** and **61** regardless of the operation mode. This means that the distortion characteristic of the power amplifier circuits **11** to **20** can be improved regardless of the operation mode.

(193) The impedance circuits **501** and **601** are not coupled between the first-stage transistor element **71** and the final-stage transistor elements **51** and **61**, and as a result, it is possible to reduce the likelihood of a mismatch of the input of the final-stage transistor elements **51** and **61**.

Consequently, in the power amplifier circuit **18** including amplifiers in two stages, it is possible to suppress the degradation of the input-output characteristics of the final-stage transistor elements **51** and **61** due to the impedance circuits **501** and **601**.

(194) The embodiments described above have been made for ease of understanding the present disclosure and should not be construed to limit the present disclosure. The present disclosure can be changed or improved without departing from its spirit and also includes equivalents thereof. This means that modifications to the embodiments designed by those skilled in the art as appropriate are also embodied in the scope of the present disclosure when the modifications have features of the present disclosure. For example, the elements included in the embodiments, the arrangements thereof, materials, conditions, shapes, sizes are not limited to the examples and may be changed as appropriate. The embodiments described above are mere examples, and as might be expected, the configurations described in the different embodiments may be partially replaced or combined with each other. These modifications are embraced within the scope of the present disclosure when these modifications contain the attributes of the present disclosure.

Claims

1. A power amplifier circuit comprising: a first transistor having a base coupled to an input terminal, a collector coupled to an output terminal, and an emitter coupled to ground; a first resistance element; a first bias circuit coupled to the base of the first transistor via the first resistance element, the first bias circuit being configured to supply a bias to the first transistor; a second transistor having a base coupled to the input terminal, a collector coupled to the output terminal, and an emitter coupled to ground; a second resistance element; a second bias circuit coupled to the base of the second transistor via the second resistance element, the second bias circuit being configured to supply a bias to the second transistor; and a first impedance circuit having a first end coupled between the base of the first transistor and the input terminal, and a second end coupled between the first bias circuit and the first resistance element, the first impedance circuit being configured to have an open state for a direct-current component of a signal passing through the first impedance circuit, and have a closed state for an alternating-current component of the signal passing through the first impedance circuit, wherein the first end of the first impedance circuit is coupled to a first node, the first node being on a first wire connecting the input terminal and the base of the first transistor, and wherein a distance of the first wire from the input terminal to the first node is more than 0.5 times a distance of the first wire from the input terminal to the base of the first transistor.

2. The power amplifier circuit according to claim 1, further comprising: a second impedance circuit having a first end coupled between the input terminal and the base of the second transistor, and a second end coupled between the second bias circuit and the second resistance element, the second

impedance circuit being configured to have an open state for a direct-current component of a signal passing through the second impedance circuit, and to have a closed state for an alternating-current component of the signal passing through the second impedance circuit.

3. The power amplifier circuit according to claim 2, wherein: the power amplifier circuit comprises one or more sets of the first resistance element and the first transistor, the power amplifier circuit comprises one or more sets of the second resistance element and the second transistor, the first impedance circuit comprises a first capacitor, the second impedance circuit comprises a second capacitor, there are more sets of the first resistance element and the first transistor than sets of the second resistance element and the second transistor, and a capacitance of the first capacitor is greater than a capacitance of the second capacitor.

4. The power amplifier circuit according to claim 2, wherein: the power amplifier circuit comprises one or more sets of the first resistance element and the first transistor, the power amplifier circuit comprises one or more sets of the second resistance element and the second transistor, the first impedance circuit comprises a first capacitor, the second impedance circuit comprises a second capacitor, there are an equal number or fewer sets of the first resistance element and the first transistor than sets of the second resistance element and the second transistor, and a capacitance of the first capacitor is equal to or less than a capacitance of the second capacitor.

5. The power amplifier circuit according to claim 2, wherein: the power amplifier circuit comprises one or more sets of the first resistance element and the first transistor, the power amplifier circuit comprises one or more second sets of the second resistance element and the second transistor, the first impedance circuit comprises a first capacitor, the second impedance circuit comprises a second capacitor, and a ratio of a capacitance of the first capacitor to a capacitance of the second capacitor is between 0.75 and 1.25 times larger than a ratio of a number of sets of the first resistance element and the first transistor to a number of sets of the second resistance element and the second transistor.

6. The power amplifier circuit according to claim 2, further comprising: a first matching circuit having a first end coupled to the input terminal, the first end of the first impedance circuit, and the first end of the second impedance circuit, and having a second end coupled to the base of the first transistor and the base of the second transistor.

7. The power amplifier circuit according to claim 6, wherein: the first impedance circuit comprises a first capacitor, the second impedance circuit comprises a second capacitor, and a first capacitance of the first capacitor is less than a second capacitance of the first capacitor, the second capacitance being obtained when the first impedance circuit is coupled between the first matching circuit, and the base of the first transistor and the base of the second transistor, and a third capacitance of the second capacitor is less than a fourth capacitance of the second capacitor, the fourth capacitance being obtained when the second impedance circuit is coupled between the first matching circuit, and the base of the first transistor and the base of the second transistor.

8. The power amplifier circuit according to claim 2, further comprising: a first matching circuit having a first end coupled to the input terminal and a second end coupled to the first end of the first impedance circuit, the first end of the second impedance circuit, the base of the first transistor, and the base of the second transistor.

9. The power amplifier circuit according to claim 8, wherein: the first impedance circuit comprises a first capacitor, the second impedance circuit comprises a second capacitor, and a second capacitance of the first capacitor is greater than a first capacitance of the first capacitor, the first capacitance being obtained when the first impedance circuit is coupled between the input terminal and the first matching circuit, and a fourth capacitance of the second capacitor is greater than a third capacitance of the second capacitor, the third capacitance being obtained when the second impedance circuit is coupled between the input terminal and the first matching circuit.

10. The power amplifier circuit according to claim 1, wherein the first impedance circuit comprises a first capacitor.

11. The power amplifier circuit according to claim 2, wherein the second impedance circuit comprises a second capacitor.
 12. The power amplifier circuit according to claim 10, wherein the first impedance circuit further comprises a third resistance element coupled in series with the first capacitor.
 13. The power amplifier circuit according to claim 10, wherein the first impedance circuit further comprises a first inductor coupled in series with the first capacitor.
 14. The power amplifier circuit according to claim 11, wherein the second impedance circuit further comprises a fourth resistance element coupled in series with the second capacitor.
 15. The power amplifier circuit according to claim 11, wherein the second impedance circuit further comprises a second inductor coupled in series with the second capacitor.
 16. The power amplifier circuit according to claim 2, wherein: the first end of the second impedance circuit is coupled to a second node, the second node being on a second wire connecting the input terminal and the base of the second transistor, and a distance of the second wire from the input terminal to the second node is more than 0.5 times a distance of the second wire from the input terminal to the base of the second transistor.
 17. A power amplifier circuit comprising: a first transistor having a base, a collector coupled to an output terminal, and an emitter coupled to ground; a first resistance element; a first bias circuit coupled to the base of the first transistor via the first resistance element, the first bias circuit being configured to supply a bias to the first transistor; a second transistor having a base, a collector coupled to the output terminal, and an emitter coupled to ground; a second resistance element; a second bias circuit coupled to the base of the second transistor via the second resistance element, the second bias circuit being configured to supply a bias to the second transistor; a first matching circuit; a second matching circuit; a third transistor having a base coupled to an input terminal via the second matching circuit, a collector coupled to the base of the first transistor and the base of the second transistor via the first matching circuit, and an emitter coupled to ground; a first impedance circuit having a first end coupled between the base of the third transistor and the second matching circuit, and a second end coupled between the first bias circuit and the first resistance element, the first impedance circuit being configured to have an open state for a direct-current component of a signal passing through the first impedance circuit, and to have a closed state for an alternating-current component of the signal passing through the first impedance circuit; and a second impedance circuit having a first end coupled between the base of the third transistor and the second matching circuit, and a second end coupled between the second bias circuit and the second resistance element, the second impedance circuit being configured to have an open state for a direct-current component of a signal passing through the second impedance circuit, and to have a closed state for an alternating-current component of the signal passing through the second impedance circuit.
 18. The power amplifier circuit according to claim 1, wherein the first impedance circuit is connected between the first bias circuit and the input terminal.
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